
PDS channel activity meets firmware dead time

Matched FD-VD/FD-HD LArSoft results and a
1024-versus-512 trace replay

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The simulation now connects activity to channel busy time

CURRENT READING

VD cathode channels are 3.6–4.3 times busier before firmware loss.

Replaying the actual candidate times through the fixed-record busy gate shows that 512 samples recovers activity in both technologies, especially VD.

01 Input activity

Eight shards per detector and source at a matched 0.7-PE threshold.

02 Coupled result

Actual per-channel times drive the XC fixed-record busy gate.

03 Still missing

Continuous ADC, full XC/CFD, transport, noise, and containment.

WHY THIS MATTERS

Dead time is not only a throughput metric

It changes which optical signatures survive long enough to become trigger evidence.

Low-energy and non-beam signatures stress the readout differently

LOCAL, BRIGHT, OR PROMPT

One interaction is visible

A localized flash gives a time anchor. Peak, integral, width, and multiplicity describe it; nearby activity may still overlap the busy window.

LOW-LIGHT OR DISTRIBUTED

The pattern carries the evidence

A supernova-like signal is a detector-wide rate excess. Near threshold, stable baselines and noise rejection protect many modest observations.

COMMON FAILURE MODE

Dead time removes evidence before DAQ can combine it

A lost waveform is not merely fewer bytes: it can bias local multiplicity, distort time structure, or suppress the rate excess used by a downstream non-beam trigger.

Local signatures depend on pulse-level evidence

Signature	Expected optical pattern	Preserve	Dead-time sensitivity
Prompt energetic interaction	Bright, localized flash across nearby channels; ns- μ s	leading time, peak, integral, width, saturation, multiplicity	nearby or delayed activity can fall inside the active window
Low-energy isolated interaction	Few-PE evidence across one or several channels; ns- μ s	near-threshold efficiency, timing, charge, coincidence	one lost pulse can break a coincidence or cross a threshold
Delayed activity	Secondary pulse or cluster after prompt light; μ s-ms	separation, second-pulse efficiency, peak count, total light	the second component can arrive while the channel is busy

LOCAL QUESTION

What survives around one accepted trigger?

These signatures test amplitude response, waveform containment, retrigger spacing, and the interpretation of nearby pulses.

Distributed signatures depend on rate and topology

Signature	Expected optical pattern	Preserve	Dead-time sensitivity
Supernova-like burst	Detector-wide rate excess of modest interactions; ms–s	rate, timing distribution, multiplicity, live time	rate-dependent losses can distort the burst excess
Nucleon-decay-like candidate	Prompt light with possible delayed components; ns–ms	prompt time, delayed evidence, coincidence structure	losses may remove the delayed or lower-amplitude component
Background and noise	Dark counts, radiological activity, correlated noise, after-pulses; ns–s	baseline, quality flags, classification, retrigger behavior	background creates dead time and time-dependent signal efficiency

DETECTOR QUESTION

What pattern reaches the downstream trigger?

These signatures test whether channel-level losses bias multi-channel rates, coincidences, and long-window structure.

One channel recognizes a pulse; the detector recognizes a signature

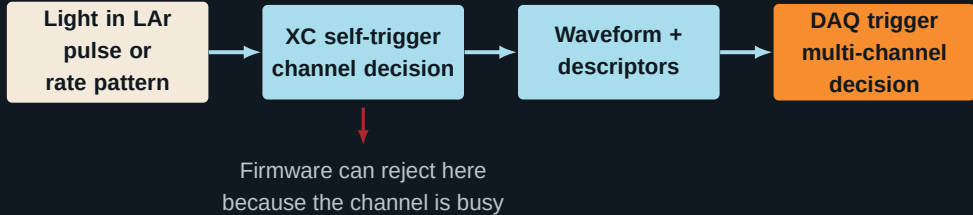


Diagram. The board exports local evidence. Detector-scale low-energy and non-beam decisions require DAQ to combine that evidence across channels, boards, and time windows.

The simulation is already useful—but its claims have boundaries

Separate waveform response, architecture dead time, and HDL signoff so one result is never asked to prove all three.

The XC emulator follows the important DAPHNE trigger stages

01 Condition

Baseline filtering and the configured pulse polarity prepare ADC samples.

02 Recognize

A 32-tap matched filter and two-cycle threshold crossing create the trigger.

03 Describe

CFD gating and CIEMAT logic produce timing, peak, integral, width, and peak count.

04 Admit

Builder, queue, ring, serializer, and output state determine whether a trigger survives.

IMPLEMENTATION STATUS

Two complementary models are in the repository

`st_xc_sim` replays waveform-level self-trigger behavior. `ring_deadtime_sim` scans architecture choices quickly; sparse HDL tests remain the signoff reference.

Each simulation answers a different question

WAVEFORM REPLAY

Will this pulse trigger and what is recorded?

Uses ADC samples, RTL-aligned XC settings, descriptors, and a selected frame length. It exposes threshold response, timing, truncation, and busy-window loss.

ARCHITECTURE SWEEP

How does dead time scale with rate?

Uses stochastic arrivals plus admission, queue, ring, serializer, and FIFO state. It compares rates cheaply, but contains no photon physics or event topology.

SIGNOFF BOUNDARY

HDL spot checks connect the two

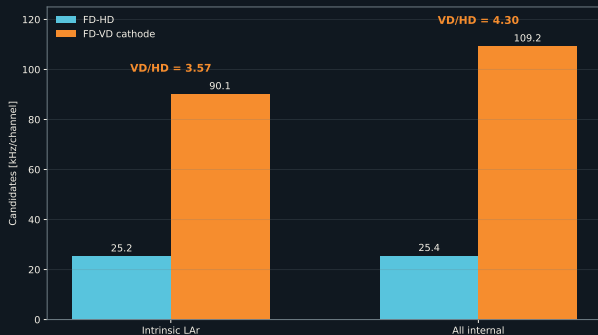
Overlay selected HDL points on the dense C++ curves. Do not label a C++ architecture sweep as firmware validation until those points agree within an accepted tolerance.

WHAT LARSOFT SAYS

One VD cathode channel is several times busier than one HD channel

The comparison holds threshold, waveform model, source category,
and hardware-channel denominator fixed.

Before firmware loss, VD cathode channels are 3.6–4.3 times busier



3.57 Intrinsic LAr

VD cathode / HD per channel.

4.30 All internal

Intrinsic plus detector materials.

NOT A GAIN FACTOR

Raw channel activity

No area correction or firmware loss.

Diagram. Matched 0.7 PE; eight shards/source; complete 6,000-channel HD and 640-channel VD-cathode denominators.

The earlier 2025 study points in the same direction

Comparison	May 2025 slides	This campaign	Reading
Intrinsic/internal light	roughly 4–5 times higher in VD	3.57–4.30	same several-times-busier scale
External gammas	roughly 20 times higher in VD	not included	boundary model still needed
Absolute rate Channel population	optical hits above 15 ADC reduced; VD classes mixed	episodes above 0.7 PE full; VD classes separate	not directly comparable denominators differ

BOTTOM LINE

There is no single factor of ten

The source-specific ratios span several to tens. For internal backgrounds, both studies support a VD/HD per-channel ratio of order four.

Source: P. Barham Alzás, “Hits and clusters,” slides 4, 6–7, May 2025.

The new coupling uses real time structure on every channel

01 LArSoft

Generate light and extract 0.7-PE threshold-candidate times.

02 Partition

Reset state independently for each event and hardware channel.

03 Replay

Apply the XC legacy fixed-record gate to the same ordered timestamps.

04 Compare

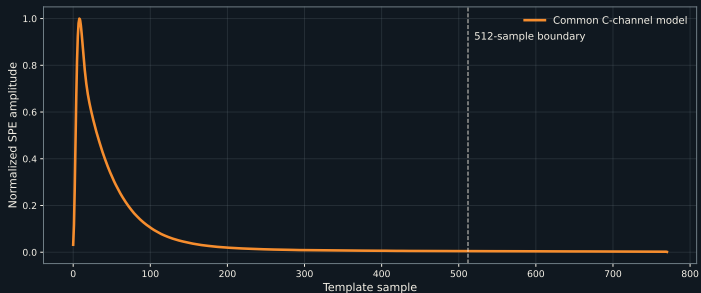
Use 1037 busy cycles for 1024 and 525 cycles for 512.

EXACTLY WHAT THIS ADDS

Candidate-level firmware acceptance

The replay preserves burstiness and cross-channel event timing while maintaining independent channel state. It does not rerun the 32-tap XC filter or CFD because continuous ADC streams were not saved.

One pinned measured shape controls the electronics comparison



SOURCE

16 Waffles cathode C channels

Large-pulse, zero-subtracted NP02 waveforms at commit 6e6274d.

CONTROLLED USE

One temporary common model

The aligned mean is used for both HD and VD; it is a shape model, not an FD calibration.

Diagram. 771 samples at 16 ns; 10 ADC/PE peak; the same strict > 7 -ADC threshold in HD and VD.

A captured waveform keeps one channel busy

The 1024-sample format protects context, but it also suppresses triggers that arrive while the builder is occupied.

Shortening the record attacks the longest first-order busy interval



Diagram. Halving the fixed record roughly halves the first-order builder occupancy; shared transport and admission rules still matter.

LEGACY MODEL

1024 samples

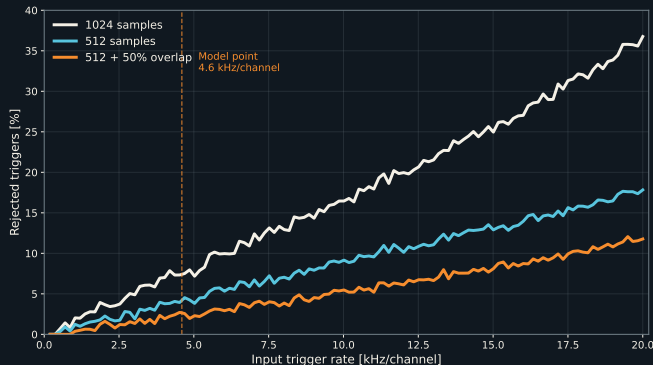
About 1037 busy cycles and 232 output words per accepted trigger.

FIRST MITIGATION

512 samples

About 525 busy cycles and 120 output words, while retaining the existing fixed-record concept.

The model says 512 samples is the largest single practical gain



4.6k Model point

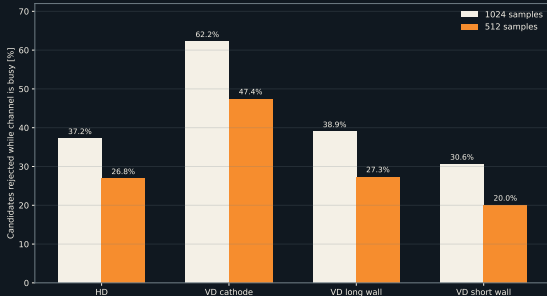
Architecture point: modeled dead time falls from 7.41% to 4.20%.

10k Stress point

16.46% becomes 9.14% with plain 512.

Diagram. Independent-Poisson sweep: at 20 kHz/ch, 36.76% becomes 17.83%. The 4.6-kHz marker is a model coordinate, not a measured detector rate.

512 improves acceptance on actual LArSoft timing



HD 37.2% → 26.8%

Intrinsic-LAr busy rejection.

VD cathode

62.2% → 47.4%

Higher activity makes length matter more.

Diagram. Same candidates and threshold; 1037 versus 525 busy cycles. VD accepted rate rises 39%, from 34.1 to 47.4 kHz/channel.

One deterministic replay cannot rank 512 against 1024

WHAT THE REPLAY ESTABLISHES

Accepted 512-sample frames contained complete pulses

For one channel and threshold, no reference pulse overlapping an accepted 512-sample frame was cut by its boundary.

WHAT IT CANNOT ESTABLISH

Raw capture counts do not rank frame lengths

Frame length changes which later triggers meet a busy channel. One fixed arrival sequence can therefore make a window look worse by phase alone.

CORRECT INTERPRETATION

Compare against common generated truth over many controlled ensembles

Scan shape, amplitude, separation, background, threshold, and seed; report containment separately from busy-path rejection.

WHAT CAN STILL GO WRONG

The simulation is coherent, but the physics envelope is incomplete

The largest uncertainty is not whether the code runs; it is whether the inputs span the signatures we care about.

Four gaps prevent a detector-level claim today

INPUT COVERAGE

Signals are not yet representative

The source ensemble excludes electronics noise, dark counts, afterpulsing, external gammas, and dedicated signal overlays.

RTL EVIDENCE

Dense curves are still C++

Sparse HDL overlays must establish agreement at low-energy, HD-anchor, and stress operating points.

MODEL COUPLING

Candidate timing is now connected

The fixed-record busy gate runs on actual LArSoft timestamps. Continuous ADC, XC/CFD, serializer, FIFO, and board transport remain separate.

PHYSICS METRIC

Dead-time fraction is not enough

We need efficiency and bias for observables used downstream: timing, integral, width, multiplicity, and burst-rate reconstruction.

Dead time can bias low-energy physics before it looks catastrophic

SINGLE CHANNEL

Acceptance loss

A second pulse inside the busy window disappears or merges with recorded activity.

LOCAL CLUSTER

Shape distortion

Multiplicity, inter-pulse spacing, and integrated light become rate dependent.

DETECTOR TRIGGER

Selection bias

A downstream burst or coincidence algorithm sees fewer and less representative primitives.

WHY NON-BEAM IS EXPOSED

There may be no external trigger to recover what firmware rejected

Beam events can have an independent time structure and trigger context. For autonomous low-energy and non-beam searches, the optical evidence may be the trigger seed itself.

Move from candidate replay to continuous waveform re- play

The present result establishes the fixed-record effect; it does not select a final software or firmware architecture.

One possible campaign would connect four controlled layers

01 Define light

Export photon-arrival ensembles for prompt, delayed, burst, and background cases.

02 Shape ADC input

Convolve arrivals with measured SPE templates, gain spread, baseline, and noise.

03 Replay DAPHNE

Run XC, CFD, descriptors, frame capture, busy logic, and 1024/512 alternatives.

04 Score physics

Measure acceptance and bias in the observables available to DAQ trigger algorithms.

A STUDY DESIGN, NOT A CHOSEN ARCHITECTURE

Several implementations remain credible

LArSoft could generate or host the signal ensemble; the standalone simulator could supply carefully selected XC, descriptor, and busy-path behavior; a hybrid could exchange timestamped photons or ADC waveforms.

A dense campaign could span this controlled signal matrix

Scan axis	Possible coverage
Response family	common C-channel baseline; then distinct validated HD and VD response families
Signal size	sub-PE/noise region through large and saturated pulses
Pulse separation	below 512 samples; near 512; between 512 and 1024; well beyond 1024
Spatial topology	single channel; neighboring channels; board-wide; multi-board
Arrival model	isolated; Poisson; clustered; delayed secondary; burst-like
Background	baseline noise; dark counts; afterpulsing; radiological and correlated activity
Firmware configuration	512 versus 1024 with identical XC, CFD, threshold, descriptor, and busy-path settings
Reported result	trigger efficiency; containment; busy rejection; timing, charge, and multiplicity bias

Interpretation. This is useful coverage, not an approved implementation; the ensemble could be standalone, LArSoft-based, or hybrid.

A denser study could report results per signature

Signature class	What firmware must preserve	Validation metric
Prompt local flash	leading time, integral, width, nearby pulses	trigger efficiency and timing/integral bias vs amplitude
Delayed local activity	second-pulse acceptance after a bright pulse	efficiency vs separation and first-pulse size
Low-light interaction	near-threshold pulse evidence across channels	efficiency vs PE, noise, baseline, and coincidence multiplicity
Detector-wide burst	rate and time structure over many channels	reconstructed excess, false-alarm rate, and dead-time bias

USEFUL COMPARISON

Hold the generated truth and operating conditions fixed

Cover HD-like and VD-cathode response families; compare 512 and 1024 with identical light, noise, threshold, busy-path logic, and random seeds.

Interpreting the result crosses three domains

PDS / PHYSICS

Signal interpretation

Representative signatures, technologies, backgrounds, rates, and observables define the physics envelope.

FIRMWARE

Implementation interpretation

RTL-aligned configuration, 1024/512 comparisons, reject counters, and HDL overlays bound the hardware claim.

DAQ / TRIGGER

Downstream interpretation

Trigger studies show how surviving descriptors and waveforms contribute to TP/TA logic and detector decisions.

WHAT WOULD STRENGTHEN THE REPORT

A reviewable evidence package

Inputs, configuration, seeds, C++ outputs, HDL checkpoints, acceptance plots, and limitations would make a 1024-versus-512 comparison reproducible.

What we know—and what remains open

SUPPORTED TODAY

512 improves modeled live acceptance

On actual LArSoft candidate times, the shorter record reduces busy rejection for HD and every VD class. Continuous-waveform truncation remains unmeasured.

OPEN QUESTION

Does the gain survive full XC/CFD replay, transport, noise, and containment requirements?

The candidate-level result motivates that test. Standalone, LArSoft-based, and hybrid implementations remain open choices.

Evidence map and claim boundaries

Evidence	Supports	Does not yet support
st_xc_sim	XC response, descriptors, frame capture	detector-wide trigger efficiency
Candidate-trace replay	per-event, per-channel fixed-record busy rejection	continuous ADC filtering, transport, or containment
C++ dead-time sweep	architecture ranking across rates	final RTL signoff or photon physics
HDL/full-chain points	tested implementation behavior	a dense all-signature efficiency surface
Waffles SPE templates	measured response shapes for selected ProtoDUNE-VD channels	universal HD/VD technology coverage

Source: [Waffles NP02 cathode templates](#)

Source: [Low-energy physics review](#)

Source: [DUNE Far Detector Technical Design Report](#)

Reproducible local artifacts

SIMULATION REPOSITORY

daphne_mezz_xc_sim

`st_xc_sim.cpp` replays waveforms.
`ring_deadtime_sim.cpp` scans architectures. Study notes document capture, dead time, ring behavior, and optimization.

KEY OUTPUTS

Current evidence

The 48-shard LArSoft replay compares 1024 and 512 on every channel. The four-way sweep separately ranks fixed and ring architectures.

REPRODUCIBILITY RULE

Record the exact branch, commit, input, and configuration with every physics claim

The presentation describes current evidence; it should not become the only record of how a result was produced.